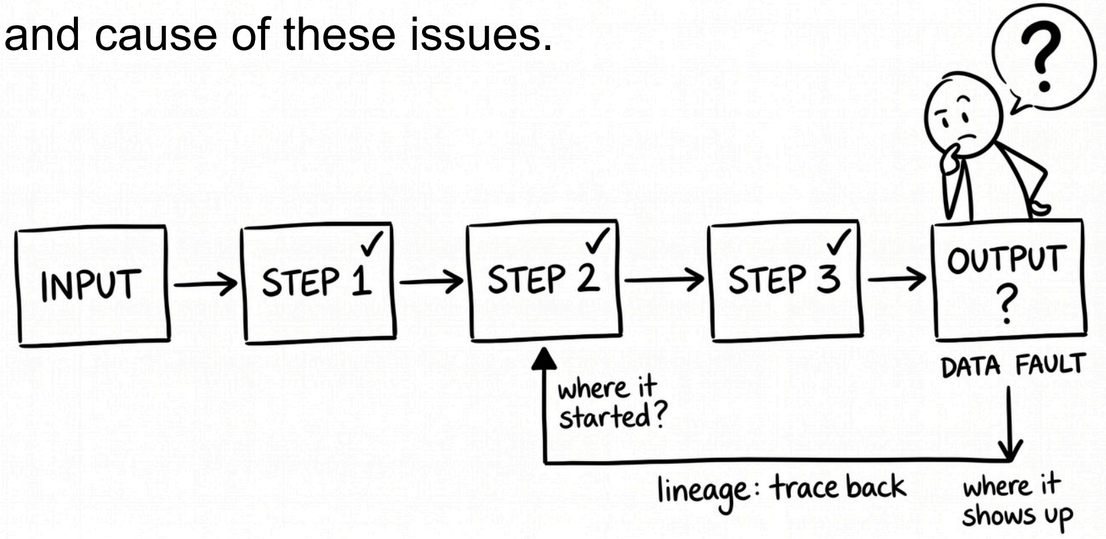
Lineage-Aware Root-Cause Analysis for Data Engineering Pipelines

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Summary: **Data pipelines often fail** silently, this means that every step reports success, but the data is wrong. Spotting that something is wrong is well supported but finding **where** it started, and **why**, is not.
This work **measures how much** the pipeline's **lineage** helps a language model find the origin and cause of these issues.

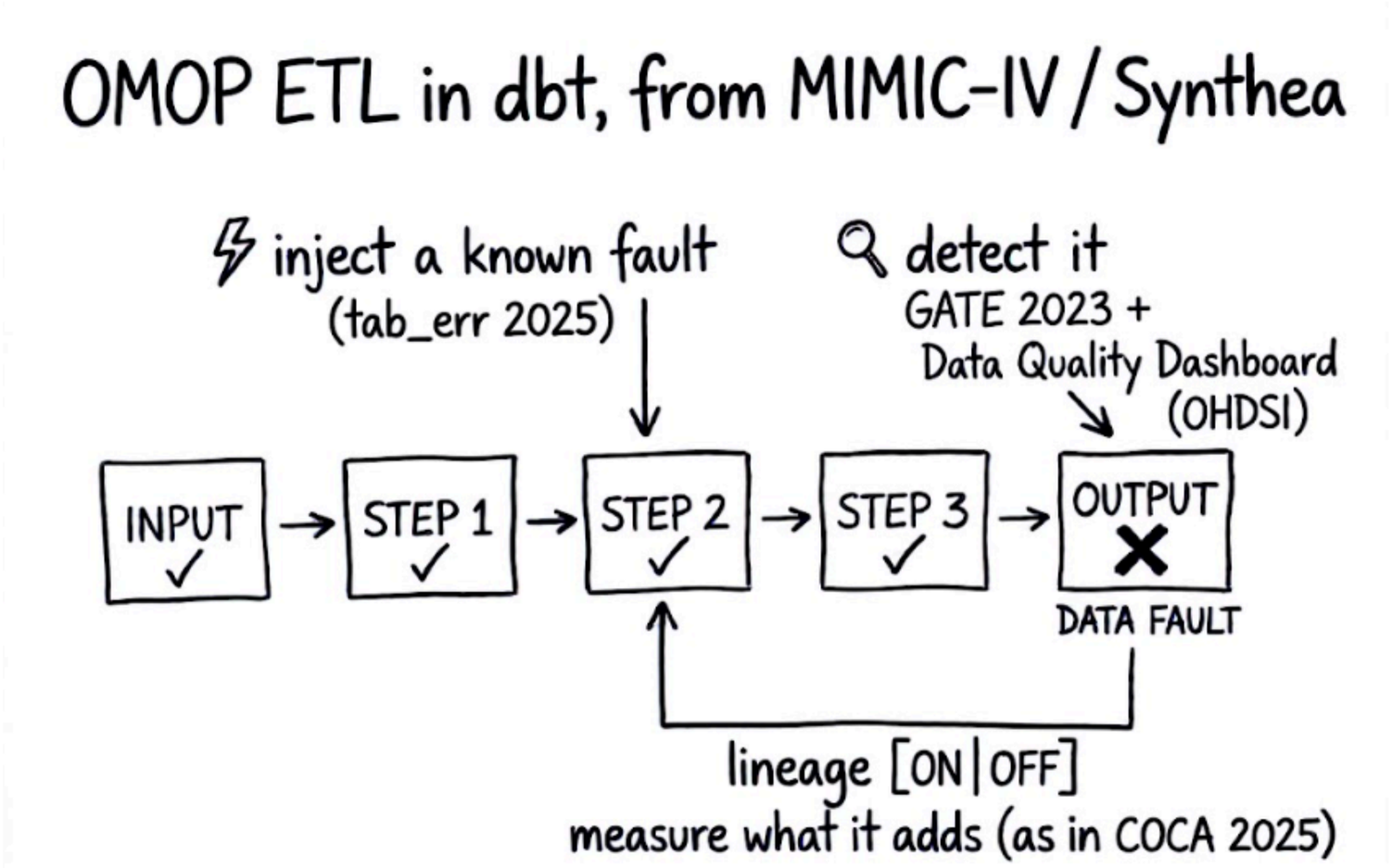


Objectives / Work done / Results

- **Literature review on three strands**: data pipelines, RCA & lineage, OMOP CDM
- **Scope**: 2021–2026, peer-reviewed papers from Scopus or Web of Science-indexed venues.
- **Search**: structured searches in SciSpace and Claude Research, with verification at source.
- **Result**: 182 returned → 122 kept → 59 relevant → 16 core

Paper	LLM and Lineage on Data Pipelines	Measurement of Lineage Effect	Known and Injected Faults
DataLineage RCA, 2025	✓	✗	✓
Data Canvas, 2026	✓	✗	✗
PRISM*, 2025	✓	✗	✓
COCA, 2025	✗	✓	✗
This work	✓	✓	✓

Next Steps / Challenges / Bottlenecks



What we take from the State of Art

- **Inject Faults:** known cause, including the wrong-but-valid mappings
- **Detection Layer**: fixed rules (DQD) + run-to-run (GATE)
- **Measure of Lineage**: per fault class and model size